Exposure Category by Gender: Followup Analysis

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Table 1. Distribution of Exposure Categories by Gender

Exposure Category	**F**	**M**	**Total**	**P-value**
Combination	20 (41%)	29 (59%)	49	0.033
Analgesic	16 (64%)	9 (36%)	25	
Unknown	13 (52%)	12 (48%)	25	
Antidepressants	17 (74%)	6 (26%)	23	
Street Drugs	9 (39%)	14 (61%)	23	
CO, As, CN	7 (32%)	15 (68%)	22	
Sedatives	7 (64%)	4 (36%)	11	
Alcohol	3 (30%)	7 (70%)	10	
Total	**92 (49%)**	**96 (51%)**	**188**	

Table 1. Counts and row-wise percentages for each exposure category by gender, including totals. Percentages in the F and M columns reflect the proportion within each row. Chi-square test found a significant association between gender and exposure category ($p = 0.033$). Categories are ordered by decreasing total count. ‘Other’ category was removed due to small sample size, but ‘Unknown’ was retained.

Table 2. Focused Comparisons: High-Frequency Categories vs All Others

Antidepressants vs All Others

	F	M	Total	P-value
All Others	75 (45%)	90 (55%)	165	0.020
Antidepressants	17 (74%)	6 (26%)	23	

Analgesic vs All Others

	F	M	Total	P-value
All Others	76 (47%)	87 (53%)	163	0.161
Analgesic	16 (64%)	9 (36%)	25	

Street Drugs vs All Others

	F	M	Total	P-value
All Others	83 (50%)	82 (50%)	165	0.434
Street Drugs	9 (39%)	14 (61%)	23	

Table 2. Binary comparisons examining whether high-frequency exposure categories show different gender distributions compared to all other categories combined. Antidepressants vs all others: chi-square test $p = 0.020$ (significant). Analgesic vs all others: chi-square test $p = 0.161$ (not significant). Street Drugs vs all others: chi-square test $p = 0.434$ (not significant).

Visualizing the Data

Figure S1. Raw Counts of Exposure Categories by Gender

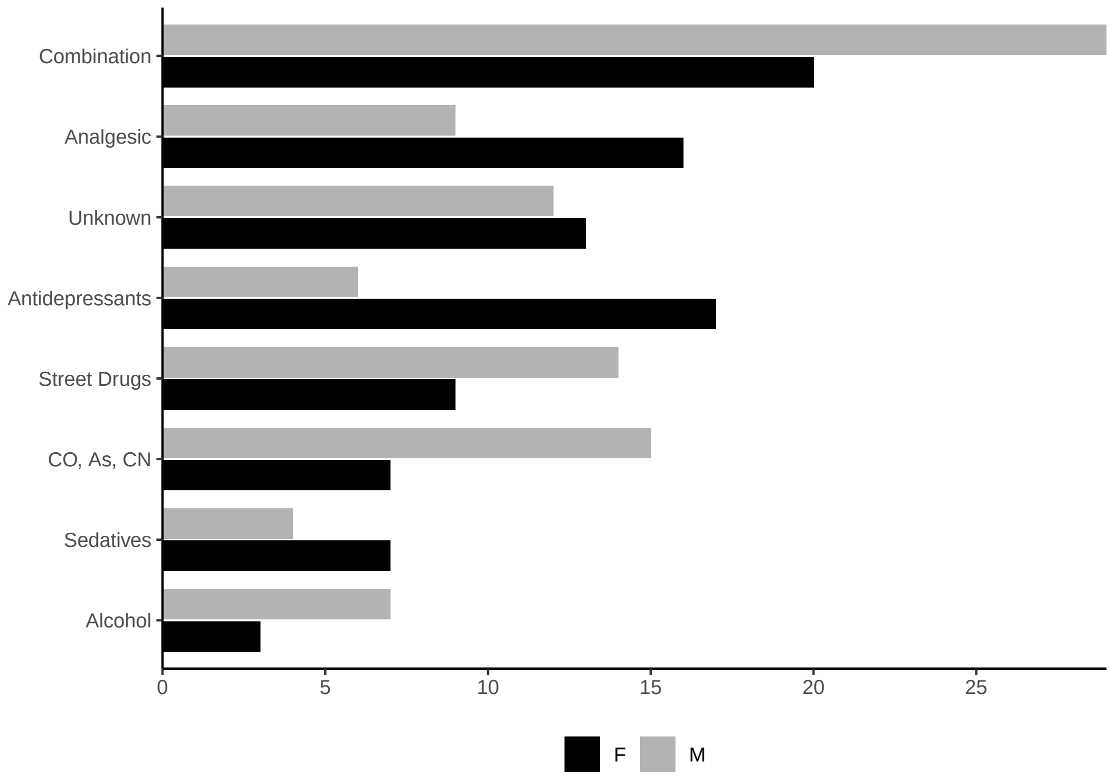

Figure S1: Raw counts of individuals in each exposure category by gender, ordered by decreasing total count to match Table 1. Females show higher counts in antidepressants and analgesics, whereas males are more represented in CO/As/CN and street drug exposures. Chi-square test shows a significant overall association.

Figure S2. Gender Distribution Within Exposure Categories (in Descending Order for Females)

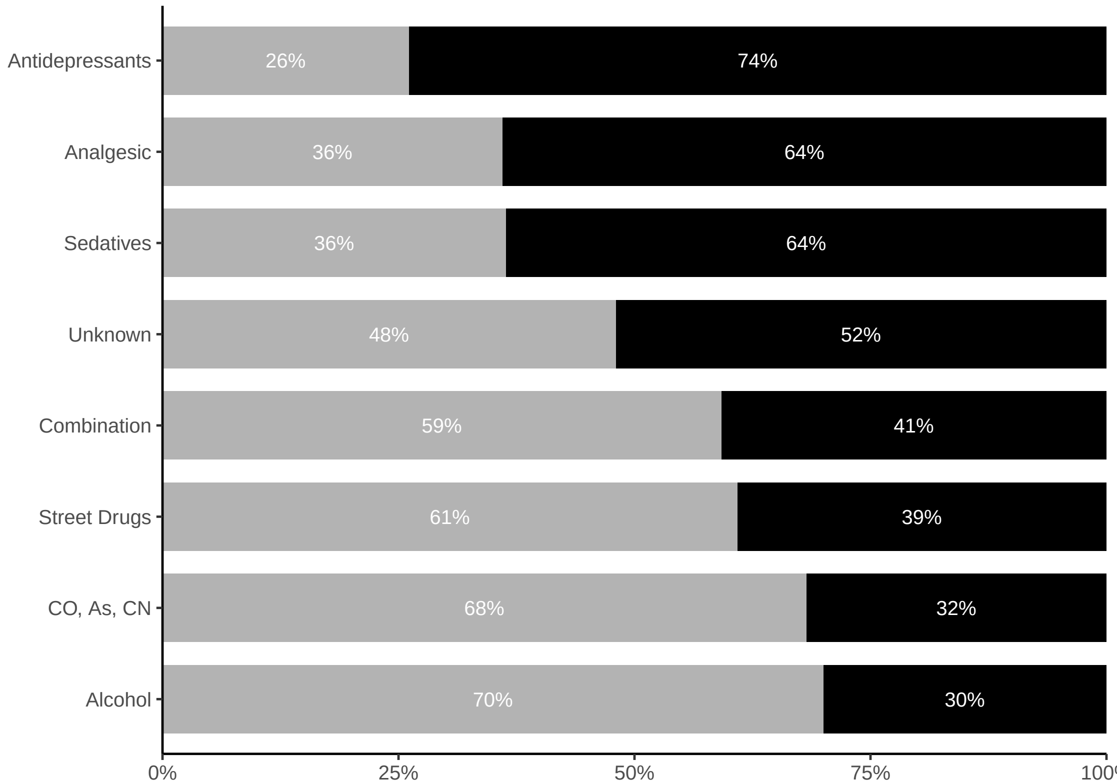

Figure S2: Gender distribution within each exposure category (each bar sums to 100%), ordered in descending order for females. Antidepressants and analgesics show the highest female representation, while CO/As/CN and alcohol show male predominance. Chi-square test indicates a significant association (p = 0.033).

Figure S3. Gender Distribution Within Exposure Categories (in Descending Order for Males)

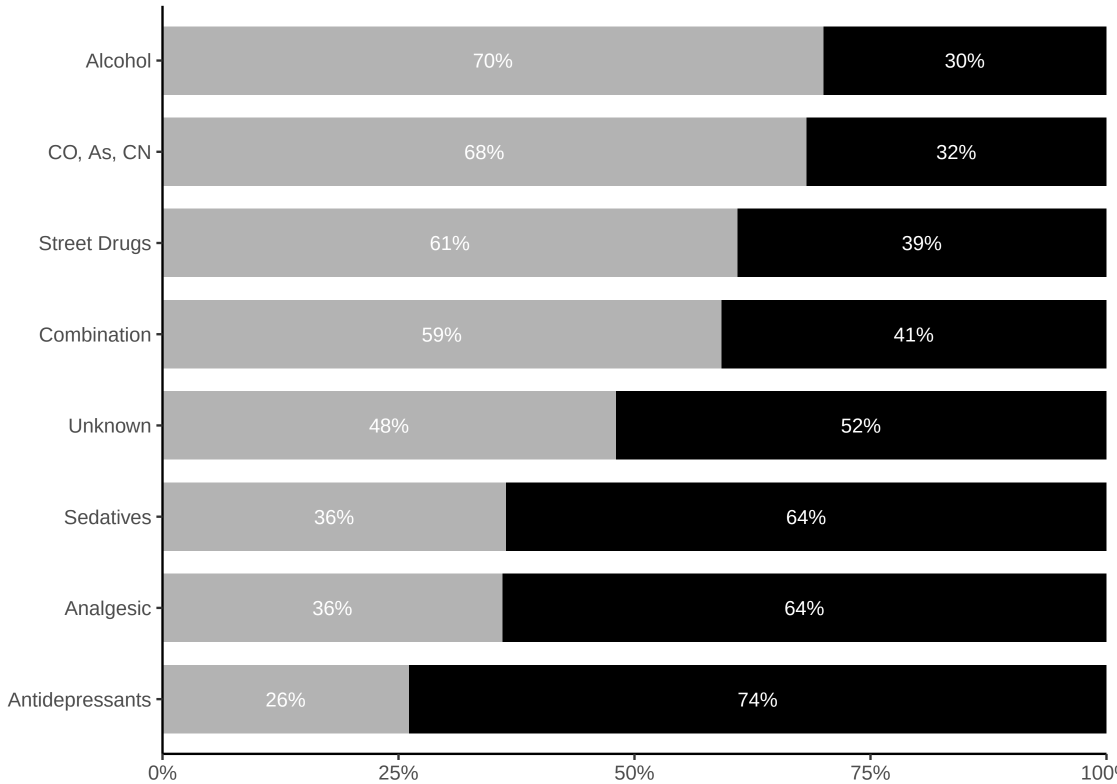

Figure S3: Gender distribution within each exposure category (each bar sums to 100%), ordered in descending order for males. Alcohol and CO/As/CN show the highest male representation, while antidepressants and analgesics show female predominance. This alternate ordering highlights categories with male predominance at the top of the visualization.

Interpretation

This followup analysis examined gender as a potential confounding variable in toxicology consultations, building on the original INTOXICATE validation study. The sample included 92 females (49%) and 96 males (51%) after removing the "Other" category due to insufficient sample size (Unknown was retained).

Statistical Analysis Results:

Chi-square test found **a significant association** between gender and exposure category ($p = 0.033$). Focused comparisons of high-frequency categories showed:

- Antidepressants vs all others: $p = 0.020$ (significant)
- Analgesic vs all others: $p = 0.161$ (not significant)
- Street Drugs vs all others: $p = 0.434$ (not significant)

Key Findings:

- **Gender Distribution Patterns:** Females comprised the majority of antidepressant and analgesic exposures, while males represented the majority of CO/As/CN and street drug exposures.

- **Clinical Significance:** The significant statistical results indicate that the observed gender patterns reflect true population associations rather than sampling variation. This suggests gender-based differences in exposure patterns that warrant consideration in clinical decision-making.
- **Methodological Improvements:** Removing low-frequency categories improved statistical power, while Fisher's exact test provided more appropriate analysis for the observed cell sizes compared to chi-square methods.

Implications for Original Study:

The significant gender associations in this toxicology consultation dataset indicate that gender should be considered as a confounding variable in the original INTOXICATE model validation. This suggests that gender-based stratification could improve model performance and that gender effects warrant investigation in larger validation studies.

Study Limitations:

The analysis focused on toxicology consultations rather than all emergency department presentations, potentially limiting generalizability. Additionally, the binary gender classification may not capture the full spectrum of gender identity in clinical populations.

Conclusion:

The statistical analysis demonstrates that observed gender patterns across exposure categories represent meaningful associations rather than random variation. These findings support considering gender as a confounding variable in clinical decision-making tools like INTOXICATE.